Data Analytics

Exercise 4: Joins, union, filtering and aggregates.

1. Retrieve all employees and their assigned projects using an inner join.
- Return EmployeeID, FirstName, LastName, Department, Salary, ProjectID, ProjectName, Budget, Status.

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary,
       ProjectID,
       ProjectName,
       Budget,
       Status
From employees AS A
INNER JOIN projects AS B
ON A.EmployeeID = B.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Complete
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Complete
6	Michael	Clark	Finance	80000	106	Risk Management	80000	pending

2. Retrieve all employees and their assigned projects, including employees who have no projects using a left join.
- Return EmployeeID, FirstName, LastName, Department, Salary, ProjectID, ProjectName Budget, Status.

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary,
       ProjectID,
       ProjectName,
       Budget,
       Status
From employees AS A
LEFT JOIN projects AS B
ON A.EmployeeID = B.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	pending

3. Retrieve all projects and their assigned employees, including projects that have no employees using a RIGHT JOIN:

- Return: ProjectID, ProjectName, Budget, Status, EmployeeID, FirstName, LastName, Department, Salary.

```
SELECT ProjectID,
       ProjectName,
       Budget,
       Status,
       EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary
From employees AS A
RIGHT JOIN projects AS B
ON A.EmployeeID = B.EmployeeID;
```

ProjectID	ProjectName	Budget	Status	EmployeeID	FirstName	LastName	Department	Salary
101	AI Development	100000	Completed	1	John	Doe	IT	70000
102	Employee Training	50000	Ongoing	2	Alice	Smith	HR	60000
103	Cybersecurity Audit	75000	pending	1	John	Doe	IT	70000
104	Financial Analysis	90000	Ongoing	3	Bob	Johnson	Finance	75000
105	Market Expansion	65000	Completed	5	Emma	Wilson	Sales	65000
106	Risk Management	80000	pending	6	Michael	Clark	Finance	80000

4. Retrieve all employees and projects, including those without a match in either table using a FULL OUTER JOIN
- Return: EmployeeID, FirstName, LastName, Department, Salary, projectID, projectName, Budget, Status.

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary,
       projectID,
       projectName,
       Budget,
       Salary
From employees AS A
FULL OUTER JOIN projects AS B
ON A.EmployeeID = B.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Salary
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	pending

5. Retrieve a list of all unique cities where employees are located and project statuses.
- Return: Location (Rename the column to Location using an alias)

```
SELECT City AS Location
From employees
UNION
SELECT status AS Location
From projects;
```

Location
New York
Los Angeles
Toronto
London
Sydney
Completed
Ongoing
pending

6 Retrieve a list of all cities where employees are located and project statuses, allowing duplicates:
- Return: Location (Rename the column to Location using an alias)

```
SELECT City AS Location
FROM employees
UNION ALL
SELECT Status AS Location
FROM projects;
```

Location
New York
Los Angeles
Toronto
London
Sydney
New York
Completed
Ongoing
pending
Ongoing
Completed
pending

7. Retrieve employees who earn more than 70,000
- Return EmployeeID, FirstName, LastName, Department, Salary.

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary
FROM   Employees
WHERE  Salary > 70 000;
```

EmployeeID	FirstName	LastName	Department	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

8. Retrieve employees working in either IT or Finance departments
- Return: EmployeeID, FirstName, LastName, Department, Salary

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary
FROM   Employees
WHERE  Department IN ('IT', 'Finance');
```

EmployeeID	FirstName	LastName	Department	Salary
1	John	Doe	IT	70000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

9. Retrieve projects that are not yet completed
- Return: projectID, projectName, Budget, Status.

```
SELECT projectID,
       projectName,
       Budget,
       Status
FROM   projects
WHERE  Status !='Completed';
```

ProjectID	ProjectName	Budget	Status
102	Employee Training	50000	Ongoing
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

10. Retrieve projects that have a budget greater than 70000 and are not completed.
- Return: projectID, projectName, Budget, Status

```
SELECT projectID,
       projectName,
       Budget,
       Status
FROM   projects
WHERE  Budget > 70000 AND Status != 'completed';
```

projectID	projectName	Budget	Status
103	Cybersecurity Audit	75000	pending
104	Financial Analysis	90000	ongoing
106	Risk Management	80000	pending

11. Retrieve employees from New York OR Toronto, ordered by Salary in descending order.
- Return: EmployeeID, FirstName, LastName, Department, Salary, City.

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary,
       City
From   Employees
WHERE  City IN('New York', 'Toronto')
ORDER BY Salary Desc;
```

EmployeeID	FirstName	LastName	Department	Salary	City
6	Michael	Clark	Finance	80000	New York
3	Bob	Johnson	Finance	75000	Toronto
1	John	Doe	IT	70000	New York

12. Retrieve the top 3 highest paid employees.
- Return: EmployeeID, FirstName, LastName, Department, Salary.

```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary
FROM   Employees
ORDER BY Salary DESC
LIMIT 3;
```

EmployeeID	FirstName	LastName	Department	Salary
6	Michael	Clark	Finance	80000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000

13. Find the total salary per department, sorted in descending order.
- Return: Department, TotalSalary (Rename SUM(salary) as TotalSalary).

```
SELECT SUM(salary) AS totalsalary,
       Department
FROM   Employees
GROUP BY Department
ORDER BY totalsalary DESC;
```

totalsalary	Department
165000	Finance
142000	IT
65000	Sales
60000	HR

14. Find the average salary per city, but only include cities where the average salary is greater than 65000
- Return: City, AverageSalary (Rename Avg(salary) as AverageSalary).

```
SELECT  City,
        AVG(salary) AS AverageSalary
FROM    Employees
Group by City
Having  AVG(salary) > 65000;
```

City	Average Salary
New York	75000
Toronto	75000
London	72000

15. Count the number of employees per department, including only departments with more than 1 employee.
- Return: Department, EmployeeCount (Rename Count (EmployeeID) as Employee Count).

```
SELECT  Department,
        Count(EmployeeID) AS Employee Count
FROM    Employees
Group by Department
Having  Count(EmployeeID) > 1;
```

Department	EmployeeCount
IT	2
Finance	2

16. Retrieve the number of projects per status, but only include statuses with at least 2 projects.
- Return: Status, projectCount (Rename count (ProjectID) as projectcount).

```
SELECT Status,
       Count(projectID) AS Projectcount
FROM   projects
Group by Status
Having count(projectID) >= 2;
```

Status	Projectcount
Completed	2
Ongoing	2
pending	2

17. Retrieve the total project budget per employee, but only for employees who are managing projects worth more than 150 000.
- Return: EmployeeID, FirstName, LastName, TotalProjectBudget (Rename SUM(Budget) as TotalprojectBudget.

```
SELECT EmployeeID,
       FirstName,
       LastName,
       SUM(Budget) AS TotalprojectBudget
FROM   Employees AS A
INNER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID
Group by EmployeeID, FirstName, LastName
Having SUM(Budget) > 150000;
```

Employee ID	FirstName	LastName	TotalprojectBudget
1	John	Doe	175000